*.pcp

Purpose

A `*.pcp` file holds the measured precipitation time series for a single gauge. One file per station. The filename is listed in `pcp.cli` and the same filename string is matched as the gauge key referenced from `pgage` in `weather-sta.cli`. The reader supports either daily values or subdaily values at a fixed time step.

Source

- Reader: `src/cli_pmeas.f90`
- Type definition: `src/climate_module.f90`, `type climate_measured_data` (the `pcp` array)

Format

The reader uses list-directed reads (free format, whitespace-separated). The first two lines are skipped. The third line is the station header. From line four to end of file are the time-step records.

Line 1: title

Single string, skipped.

Line 2: column header

Skipped. SWAT+ Editor writes `NBYR  TSTEP  LAT  LONG  ELEV`.

Line 3: station header

Read by `read (108,*,iostat=eof) pcp(i)%nbyr, pcp(i)%tstep, pcp(i)%lat, pcp(i)%long, pcp(i)%elev`.

#	Field	Type	Units	Description
1	nbyr	int	years	Number of years of record in the file. Used to size the in-memory time-series array.
2	tstep	int	minutes	Time step. `0` means daily. A positive value is the subdaily interval in minutes and must match `time%step` from `time.sim`.
3	lat	real	degrees	Latitude of the gauge.
4	long	real	degrees	Longitude of the gauge.
5	elev	real	m	Elevation of the gauge.

Records (line 4 onward)

The record layout depends on `tstep`.

For daily records (`tstep = 0`):

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year of the record.
2	jday	int	day-of-year	Julian day (1 to 365 or 366).

#	Field	Type	Units	Description
3	pcp	real	mm	Daily precipitation depth. Use a value at or below `-97` to flag missing data; the reader replaces it with a generated value.

For subdaily records (`tstep > 0`):

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year.
2	jday	int	day-of-year	Julian day.
3	mo	int	month	Month number.
4	day_mo	int	day	Day of the month.
5	ihr	int	step index	Sub-day step index (1 to `1440 / tstep`).
6	pcp	real	mm	Precipitation depth for the time step.

Records are read in order. The reader advances year-by-year using the change in `jday` (it expects the year to change after `jday = 365` or `366`).

Missing data

The daily reader treats any value at or below `-97` as missing. Downstream in `cli_precip_control.f90`, when `precip_next <= -97.` the value is replaced by the weather generator. The classic flag written by SWAT+ Editor and the legacy tools is `-99`.

Example

First lines of `refdata/Ames_sub1/ames.pcp`:

```
AME.pcp
 NBYR    TSTEP       LAT      LONG      ELEV
   121        0     42.04    -93.89    316
1900 1 -99
1900 2 -99
1900 3 -99
...
1983 184 0
1983 185 15.5
1983 186 0
```

Station header: 121 years, daily time step (`0`), latitude 42.04, longitude -93.89, elevation 316 m. Early years are missing (`-99`) and will be filled by the weather generator.

Related files

- `pcp.cli`. Lists each `*.pcp` filename.
- `weather-sta.cli`. Binds a precipitation gauge to a weather station via `pgage`.
- `file.cio`. Names the `pcp.cli` index and the optional path prefix `in_path_pcp%pcp` applied to every `*.pcp` filename.
- `time.sim`. For subdaily files, `time%step` must equal the file's `tstep`.